Three concurrent lines

Animath — Stage olympique de Sophia-Antipolis · 2006 · France

Let ABC be a triangle. Let H_A be the foot of the altitude from A, $\mathcal{C}_A$ the circle with diameter $[AH_A]$, and M and N the intersections of this circle with the sides of the triangle. Let L_A be the line through A perpendicular to (MN). The lines L_B and L_C are defined in the same way. Show that the three lines L_A, L_B and L_C are concurrent.